Matrix functions and stability

Algebraicity of joint spectral radii from rational input

Difficulty: extreme **Importance:** interesting to the community

Status: open; book question with subsequent-literature screening

Context and notation

The joint spectral radius of a nonempty compact set $\mathcal{M} \subset \mathbb{C}^{d\times d}$ is

$$\hat{\rho}(\mathcal{M}) = \lim_{k\to\infty} \max_{A_1,\dots,A_k\in\mathcal{M}} \|A_k \cdots A_1\|_2^{1/k}.$$

This definition also applies to finite real matrix sets. The ordinary spectral radius of one matrix is written $\rho(A)$.

Problem statement

Is it true that, for every $d \geq 1$ and every finite nonempty $\mathcal{M} \subset \mathbb{Q}^{d\times d}$, there exists a nonzero polynomial $p \in \mathbb{Z}[t]$ such that

$$p(\hat{\rho}(\mathcal{M})) = 0?$$

The claim is about the value being algebraic; it does not assert a uniform procedure for finding its minimal polynomial.

Reference and status evidence

Jungers, *The Joint Spectral Radius: Theory and Applications*, §4.4, Open Question 7, printed p. 75 of the author manuscript. Searches combining “joint spectral radius”, “rational matrices”, “algebraic number”, “algebraicity”, and “transcendental” located no proof or rational-input counterexample. The latest explicit formulation located remains the book.

Scope

Both signs are permitted. A finite-product formula would imply algebraicity for its particular input, but this question requires no such formula. It concerns exact representation of a matrix computation’s output, separately from the geometry of the set of all stable inputs.